

Characterize the optimal equilibrium stopping region under general aggregation preferences

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Problem

We are given a one-dimensional time-homogeneous diffusion X with continuous paths and the strong Markov property, a measurable payoff g , an independent nonnegative random discount rate ρ , and a strictly increasing aggregation function φ . For a Borel stopping region $R \subseteq \mathbb{R}$,

$$\tau(x, R) := \inf\{t > 0 : X_t^x \in R\}, \quad J(x, R) := \int_0^\infty \varphi\left(\mathbb{E}\left[e^{-r\tau(x, R)} g(X_{\tau(x, R)}^x)\right]\right) dF_\rho(r),$$

and

$$V(x, R) := \varphi(g(x)) \vee J(x, R).$$

A mild equilibrium R is a fixed point of the policy-improvement operator Θ , equivalently a stopping region consistent with the comparison between immediate stopping and continuation.

The broad reading of the request is:

For *arbitrary* one-dimensional diffusions, measurable g , strictly increasing φ , and discount distributions F_ρ , characterize an optimal equilibrium $R^* \in \mathcal{E}$ whenever it exists, without imposing the paper's structural condition $x \mapsto x\varphi'(x)$ increasing.

The negation of any blanket existence theorem is:

There exists an admissible model with $\mathcal{E} \neq \emptyset$ but with no $R^* \in \mathcal{E}$ such that

$$V(x, R^*) \geq V(x, R) \quad \forall x \in \mathbb{R}, \forall R \in \mathcal{E}.$$

I will prove two things:

1. a general *order-theoretic sufficient criterion* for optimality of an equilibrium which does *not* involve any condition on $x \mapsto x\varphi'(x)$, and
2. an explicit *nonexistence counterexample* showing that, without additional coercive/model assumptions, no global characterization can exist.

Alternative readings considered

- **Hardest reading (adopted):** a universal theorem for all admissible quadruples (X, g, φ, F_ρ) . This is false; I disprove it by an explicit Brownian example.
- **Weaker reading:** give checkable sufficient conditions that identify the optimal equilibrium when one exists. I do this via an extremal-equilibrium criterion.
- **Convention on $\tau = \infty$:** in the counterexample, the only place $\tau = \infty$ matters is to rule out $R = \emptyset$. I use the standard discounted-stopping convention that if $\rho > 0$ a.s., then continuation with $\tau = \infty$ yields payoff 0.

Related literature found

- **Deng–Yu–Zhang (2025) — PARTIAL.** Exact problem class: aggregation over heterogeneous discount rates with a one-dimensional diffusion. They prove sufficient conditions under which the *smallest* mild equilibrium is optimal, but not a no-condition characterization beyond those assumptions.
- **Huang–Nguyen-Huu (2018) — BACKGROUND.** Fixed-point iteration and equilibrium stopping under decreasing impatience; useful for orienting the equilibrium/fixed-point viewpoint.
- **Huang–Yu (2021) — PARTIAL.** Closely related nonlinear-aggregation/time-inconsistency model (ambiguity attitude). Useful analogy for how nonlinear aggregation changes equilibrium structure.
- **Bayraktar–Wang–Zhou (2023) — BACKGROUND.** Clarifies mild/weak/strong equilibrium notions for one-dimensional diffusions and their relations.
- **Christensen–Lindensjö (2018) — BACKGROUND.** Foundational continuous-time equilibrium formulation for time-inconsistent stopping.
- **Christensen–Klein–Schultz (2025) — BACKGROUND.** Another diffusion stopping problem made time-inconsistent by a nonstandard constraint; useful as a comparison point for what can and cannot be characterized.

None of the consulted sources gives a full characterization for the present problem class *without* additional structure. I do *not* import any proof step from those papers below.

Proof sketch

There are two logically separate issues: *optimality among equilibria* and *existence of an extremal equilibrium*.

First, I isolate the real order-theoretic mechanism behind many “smallest equilibrium is optimal” statements. If equilibria are ordered by inclusion and the continuation payoff $J(x, \cdot)$ is monotone along that order, then the least or greatest equilibrium is automatically optimal by a three-case pointwise comparison. This argument is completely independent of any derivative condition on φ ; it uses only the mild-equilibrium sign conditions.

Second, I show that no theorem of the form “an optimal equilibrium always exists” can hold in full generality. The counterexample uses a genuine diffusion: standard Brownian motion, payoff $g(x) = e^x$, linear attitude $\varphi(y) = y$, and a nondegenerate discount distribution supported in $(0, \frac{1}{2})$. For every threshold $R_a = [a, \infty)$, the discounted expected payoff from waiting until level a is

$$e^a \mathbb{E}_x[e^{-\rho\tau_a}] = \int e^{x+(1-\sqrt{2r})(a-x)} dF_\rho(r),$$

which strictly exceeds the immediate payoff e^x whenever $x < a$. Hence every threshold is an equilibrium, and higher thresholds strictly improve the value at states below them. Therefore there is an infinite improving chain of equilibria and no optimal equilibrium.

So the strongest rigorous conclusion I can support is:

1. a general sufficient optimality criterion that removes the paper's structural condition entirely, and
2. a sharp obstruction: without additional coercive assumptions on (X, g, F_ρ) , a global optimal equilibrium need not exist at all.

Main result

Theorem 1 (Extremal-equilibrium criterion). *Let $f(x) := \varphi(g(x))$. Assume that mild equilibrium is understood in the fixed-point sense of the policy-improvement operator; equivalently,*

$$x \in R \implies f(x) \geq J(x, R), \quad x \notin R \implies f(x) \leq J(x, R). \quad (\text{E})$$

(a) *Suppose there exists $R_- \in \mathcal{E}$ such that $R_- \subseteq R$ for every $R \in \mathcal{E}$, and suppose further that*

$$R_1, R_2 \in \mathcal{E}, \quad R_1 \subseteq R_2 \implies J(x, R_1) \geq J(x, R_2) \quad \forall x \in \mathbb{R}.$$

Then R_- is an optimal equilibrium.

(b) *Suppose there exists $R_+ \in \mathcal{E}$ such that $R \subseteq R_+$ for every $R \in \mathcal{E}$, and suppose further that*

$$R_1, R_2 \in \mathcal{E}, \quad R_1 \subseteq R_2 \implies J(x, R_1) \leq J(x, R_2) \quad \forall x \in \mathbb{R}.$$

Then R_+ is an optimal equilibrium.

Theorem 2 (Explicit nonexistence of an optimal equilibrium). *Let $X_t = x + W_t$ be standard Brownian motion, let*

$$g(x) = e^x, \quad \varphi(y) = y,$$

and let ρ be independent of X , nonconstant, and satisfy

$$0 < \rho < \frac{1}{2} \quad a.s.$$

Then:

(i) *for every $a \in \mathbb{R}$, the threshold region*

$$R_a := [a, \infty)$$

is a mild equilibrium;

(ii) *for every $x < a$,*

$$V(x, R_a) = J(x, R_a) = \int_0^\infty e^{x+(1-\sqrt{2r})(a-x)} dF_\rho(r),$$

and this quantity is strictly increasing in a ;

(iii) *consequently, $\mathcal{E} \neq \emptyset$ but $\mathcal{E}$ has no optimal equilibrium.*

Proof

I first record the equilibrium sign condition that will be used repeatedly.

Lemma 1 (Sign form of mild equilibrium). *Let $f(x) := \varphi(g(x))$. If R is a mild equilibrium in the fixed-point sense $R = \Theta(R)$, then*

$$x \in R \implies f(x) \geq J(x, R), \quad x \notin R \implies f(x) \leq J(x, R).$$

Proof. By definition of the policy-improvement operator, $\Theta(R)$ stops at states where stopping is strictly better, continues at states where continuation is strictly better, and preserves the current action at indifference states. Thus $R = \Theta(R)$ is equivalent to saying:

- if the policy stops at x (that is, $x \in R$), then stopping is not worse than continuation;
- if the policy continues at x (that is, $x \notin R$), then continuation is not worse than stopping.

That is exactly the displayed pair of inequalities. □

Proof of Theorem 1. Fix $R \in \mathcal{E}$.

Part (a). Assume $R_- \subseteq R$ for all $R \in \mathcal{E}$, and assume $J(x, R_1) \geq J(x, R_2)$ whenever $R_1 \subseteq R_2$ are equilibria.

We compare $V(x, R_-)$ and $V(x, R)$ pointwise.

- If $x \in R_-$, then $x \in R$ as well. By (E), both equilibria stop at x , so

$$V(x, R_-) = f(x) = V(x, R).$$

- If $x \in R \setminus R_-$, then $x \notin R_-$. By (E) for R_- ,

$$J(x, R_-) \geq f(x).$$

Since $x \in R$, again by (E) for R ,

$$V(x, R) = f(x).$$

Because $x \notin R_-$, we have $V(x, R_-) = J(x, R_-)$. Hence

$$V(x, R_-) = J(x, R_-) \geq f(x) = V(x, R).$$

- If $x \notin R$, then $x \notin R_-$ as well because $R_- \subseteq R$. Therefore

$$V(x, R_-) = J(x, R_-), \quad V(x, R) = J(x, R).$$

By the assumed monotonicity of J along equilibria,

$$J(x, R_-) \geq J(x, R),$$

so

$$V(x, R_-) \geq V(x, R).$$

Since these are the only three cases, $V(x, R_-) \geq V(x, R)$ for every $x \in \mathbb{R}$. As $R \in \mathcal{E}$ was arbitrary, R_- is optimal.

Part (b). Assume $R \subseteq R_+$ for all $R \in \mathcal{E}$, and assume $J(x, R_1) \leq J(x, R_2)$ whenever $R_1 \subseteq R_2$ are equilibria.

Again compare pointwise.

- If $x \notin R_+$, then $x \notin R$ as well. Hence

$$V(x, R_+) = J(x, R_+), \quad V(x, R) = J(x, R).$$

By monotonicity,

$$J(x, R) \leq J(x, R_+),$$

so

$$V(x, R) \leq V(x, R_+).$$

- If $x \in R$, then $x \in R_+$ too. Both equilibria stop at x , so

$$V(x, R) = f(x) = V(x, R_+).$$

- If $x \in R_+ \setminus R$, then by (E) for R ,

$$J(x, R) \geq f(x),$$

and by monotonicity $J(x, R) \leq J(x, R_+)$. On the other hand, by (E) for R_+ ,

$$f(x) \geq J(x, R_+).$$

Combining these inequalities gives

$$J(x, R) = J(x, R_+) = f(x).$$

Therefore

$$V(x, R) = J(x, R) = f(x) = V(x, R_+).$$

Thus $V(x, R_+) \geq V(x, R)$ for every x , and R_+ is optimal. $\square$

For the Brownian counterexample I need the first-passage Laplace transform.

Lemma 2 (Brownian first-passage transform). *Let $X_t = x + W_t$ with W standard Brownian motion, let $a \in \mathbb{R}$, and for $x \leq a$ define*

$$\tau_a := \inf\{t > 0 : X_t \geq a\}.$$

Then for every $r > 0$,

$$\mathbb{E}_x[e^{-r\tau_a}; \tau_a < \infty] = e^{-\sqrt{2r}(a-x)}.$$

Consequently $\mathbb{P}_x(\tau_a < \infty) = 1$. In the boundary case $x = a$, one has $\tau_a = 0$ a.s.

Proof. Fix $r > 0$ and set $\lambda := \sqrt{2r}$. Consider

$$M_t := \exp(\lambda X_t - rt) = \exp\left(\lambda x + \lambda W_t - \frac{\lambda^2}{2}t\right).$$

By Itô's formula,

$$dM_t = \lambda M_t dW_t,$$

so M is a martingale.

Now stop at $\tau_a \wedge t$. Since $x \leq a$, we have $X_s < a$ for $0 < s < \tau_a$ and $X_{\tau_a} = a$ on $\{\tau_a < \infty\}$ by continuity; hence

$$X_{\tau_a \wedge t} \leq a.$$

Therefore

$$0 \leq M_{\tau_a \wedge t} \leq e^{\lambda a},$$

so dominated convergence is available. Since M is a martingale,

$$e^{\lambda x} = \mathbb{E}_x[M_{\tau_a \wedge t}].$$

Letting $t \rightarrow \infty$, on $\{\tau_a < \infty\}$ one has

$$M_{\tau_a \wedge t} \rightarrow e^{\lambda a - r\tau_a},$$

while on $\{\tau_a = \infty\}$,

$$M_{\tau_a \wedge t} = e^{\lambda X_t - rt} \leq e^{\lambda a - rt} \rightarrow 0.$$

Hence

$$e^{\lambda x} = e^{\lambda a} \mathbb{E}_x[e^{-r\tau_a}; \tau_a < \infty],$$

which yields

$$\mathbb{E}_x[e^{-r\tau_a}; \tau_a < \infty] = e^{-\lambda(a-x)} = e^{-\sqrt{2r}(a-x)}.$$

Now let $r \downarrow 0$. Since $e^{-r\tau_a} \uparrow 1_{\{\tau_a < \infty\}}$, monotone convergence gives

$$\mathbb{P}_x(\tau_a < \infty) = 1.$$

Finally, if $x = a$, then for every $r > 0$,

$$\mathbb{E}_a[e^{-r\tau_a}] = 1.$$

Because $0 \leq e^{-r\tau_a} \leq 1$, this forces $e^{-r\tau_a} = 1$ a.s., hence $\tau_a = 0$ a.s. □

Proof of Theorem 2. Fix $a \in \mathbb{R}$ and write $R_a = [a, \infty)$.

Step 1: compute the continuation value for R_a .

If $x < a$, then $X_{\tau(x, R_a)} = a$ a.s. and the preceding lemma gives

$$\mathbb{E}_x[e^{-r\tau(x, R_a)} g(X_{\tau(x, R_a)})] = e^a \mathbb{E}_x[e^{-r\tau_a}] = e^{a - \sqrt{2r}(a-x)} = e^{x + (1 - \sqrt{2r})(a-x)}.$$

Since here $\varphi(y) = y$, we obtain

$$J(x, R_a) = \int_0^\infty e^{x + (1 - \sqrt{2r})(a-x)} dF_\rho(r), \quad x < a. \quad (1)$$

If $x > a$, then x is an interior point of R_a . By continuity of the path, for each ω there exists $\delta(\omega) > 0$ such that $X_t(\omega) > a$ for all $0 \leq t \leq \delta(\omega)$. Hence

$$\tau(x, R_a) = 0, \quad X_{\tau(x, R_a)} = x,$$

and so

$$J(x, R_a) = e^x.$$

If $x = a$, the lemma gives $\tau(a, R_a) = 0$ a.s., hence again

$$J(a, R_a) = e^a.$$

Therefore

$$J(x, R_a) = \begin{cases} \int_0^\infty e^{x+(1-\sqrt{2r})(a-x)} dF_\rho(r), & x < a, \\ e^x, & x \geq a. \end{cases}$$

Step 2: verify that every R_a is a mild equilibrium.

Because $0 < \rho < \frac{1}{2}$ a.s., we have $1 - \sqrt{2r} > 0$ for F_ρ -a.e. r . Hence for every $x < a$,

$$e^{x+(1-\sqrt{2r})(a-x)} > e^x \quad \text{for } F_\rho\text{-a.e. } r,$$

and after integration,

$$J(x, R_a) > e^x = \varphi(g(x)).$$

So at every $x < a$, continuation is strictly preferred.

For every $x \geq a$, we proved above that

$$J(x, R_a) = e^x = \varphi(g(x)).$$

So at every $x \geq a$, stopping is weakly optimal.

This is exactly the mild-equilibrium condition (E). Thus $R_a \in \mathcal{E}$ for every $a \in \mathbb{R}$.

Step 3: monotonicity in the threshold.

Fix $x \in \mathbb{R}$. If $a > x$, then by (1),

$$V(x, R_a) = J(x, R_a) = \int_0^\infty e^{x+(1-\sqrt{2r})(a-x)} dF_\rho(r).$$

Differentiating under the integral sign is justified because the integrand is smooth in a , positive, and for a in compact intervals admits an obvious integrable bound. Thus

$$\frac{\partial}{\partial a} V(x, R_a) = \int_0^\infty (1 - \sqrt{2r}) e^{x+(1-\sqrt{2r})(a-x)} dF_\rho(r).$$

Since $1 - \sqrt{2r} > 0$ a.s., the derivative is strictly positive. Therefore $a \mapsto V(x, R_a)$ is strictly increasing on (x, ∞) .

This proves (ii).

Step 4: no optimal equilibrium exists.

First, $\mathcal{E} \neq \emptyset$ because every threshold R_a is an equilibrium.

Now let $R \in \mathcal{E}$ be arbitrary. I claim $R \neq \emptyset$. Indeed, if $R = \emptyset$, then $\tau(x, R) = \infty$ for every x , and because $\rho > 0$ a.s., continuation yields payoff 0; hence

$$J(x, \emptyset) = 0 < e^x = \varphi(g(x)),$$

so $\emptyset$ cannot be a mild equilibrium.

Thus R contains some point x . Since R is a mild equilibrium and $x \in R$, the sign condition (E) gives

$$V(x, R) = e^x.$$

Choose any $a > x$. Then $R_a \in \mathcal{E}$, and by Step 3,

$$V(x, R_a) > e^x = V(x, R).$$

Therefore R fails to dominate the equilibrium R_a . As $R \in \mathcal{E}$ was arbitrary, no equilibrium dominates all the others.

Hence $\mathcal{E}$ has no optimal equilibrium. □

Gap to the full statement

What remains unproved is a *sharp* characterization for arbitrary one-dimensional diffusions and arbitrary increasing φ .

More precisely, I have *not* shown:

- how to verify, in full generality, the set-monotonicity property

$$R_1 \subseteq R_2 \implies J(x, R_1) \gtrless J(x, R_2),$$

which is the real order property needed in Theorem 1;

- how to prove the existence of a least or greatest equilibrium for the full Borel policy class $\mathcal{E}$;
- how to classify all models for which an optimal equilibrium exists.

What I *have* shown is that any such full characterization must include extra assumptions beyond mere monotonicity of φ : without them, optimal equilibria can fail to exist even in a very explicit Brownian model.

Self red-team

- **Counterexample attempt against Theorem 1.** I tried to break the proof by looking at x in the symmetric difference of two equilibria. The proof already separates exactly those cases: $x \in R_-$, $x \in R \setminus R_-$, and $x \notin R$. No hidden fourth case exists.
- **Boundary case $x = a$ in Theorem 2.** This is the most plausible technical objection. I checked it separately: the first-passage lemma gives $\mathbb{E}_a[e^{-r\tau_a}] = 1$, and since $0 \leq e^{-r\tau_a} \leq 1$, necessarily $\tau_a = 0$ a.s.
- **Is the Brownian example genuinely time-inconsistent?** Yes. With $\varphi = \text{id}$ and nonconstant ρ , the effective discount factor is $t \mapsto \mathbb{E}[e^{-\rho t}]$, a nonexponential mixture of exponentials, so the model is in the standard nonexponential-discount/time-inconsistent class.
- **Does Theorem 2 accidentally contradict known existence results?** No. The cited existence/optimal results in the literature impose additional structure (e.g. decreasing-impatience/log-subadditivity plus regularity/coercivity conditions, or boundary conditions on admissible equilibria). Here the exponential payoff $g(x) = e^x$ together with $0 < \rho < 1/2$ creates an escaping improving chain $R_a = [a, \infty)$, $a \rightarrow \infty$, so no extremal equilibrium exists.
- **Degenerate/boundary discount regimes checked.**
 - If $\rho = \frac{1}{2}$ a.s., then the strict inequality in Step 2 disappears and the nonexistence proof breaks.
 - If $\rho > \frac{1}{2}$ on a set of positive probability, higher thresholds need not improve the value.
 - Thus the assumption $0 < \rho < \frac{1}{2}$ a.s. is doing real work.
- **Only place the $\tau = \infty$ convention enters.** It is used solely to rule out $R = \emptyset$. Every equilibrium actually used in the construction is a threshold R_a with finite first passage from below.

The argument survived these checks.

Ideas tried

1. **Full lattice/fixed-point route.** I first tried to push a Tarski-style fixed-point argument on the policy-improvement operator Θ . *Obstruction:* the class of Borel subsets of $\mathbb{R}$ is not a complete lattice under arbitrary unions/intersections, so a bare Tarski theorem does not directly apply at the level stated in the problem. This suggested separating *existence* from *optimality*.
2. **Universal set-monotonicity of J .** I tried to prove that $R_1 \subseteq R_2$ should force a global order on $J(x, R_1)$ and $J(x, R_2)$. *Obstruction:* enlarging a stopping region makes future selves stop earlier but also at different states; with general diffusions/payoffs these two effects compete, so no universal monotonicity can be true without extra structure. This is why Theorem 1 assumes monotonicity instead of bluffing that it is automatic.
3. **Counterexample with linear payoff and nonlinear φ .** I tried several threshold computations with Brownian motion and $g(x) = x$, hoping to force crossing values by choosing φ that violates $x\varphi'(x)$ monotone. *Obstruction:* verifying the equilibrium inequalities globally became messy. *Next step suggested by this failed attempt:* work with a payoff that is an eigenfunction of the diffusion generator, so that threshold continuation values become explicit and separable.
4. **Successful route: exponential payoff eigenfunction.** Switching to $g(x) = e^x$ for Brownian motion makes the threshold continuation value explicit:

$$\mathbb{E}_x[e^{-r\tau_a}e^{X_{\tau_a}}] = e^{x+(1-\sqrt{2r})(a-x)}.$$

This produces an explicit improving chain of equilibria and gives the clean nonexistence theorem above.

Invoked results

No external theorem is used as a black box inside the proof.

The only stochastic input is the exponential Brownian martingale

$$\exp\left(\lambda W_t - \frac{\lambda^2}{2}t\right),$$

which I verify in-line by Itô's formula and then stop at $\tau_a \wedge t$. All remaining steps use only dominated convergence, monotone convergence, and elementary case distinctions.

Self-confidence

3 — I checked every step of the stated proofs carefully, and the Brownian counterexample is explicit and robust; the only reason this is not a 4 is that the full problem is broad and my positive result is intentionally framed as a sufficient criterion rather than a complete characterization.

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  title = {On Time-Consistent Equilibrium Stopping Under Aggregation of Diverse Discount Rates},
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